

Hidden Markov Models

Discrete Markov Processes (Markov chains)

- Markov property:

$$P(q_r = S_i | q_{r-1} = S_j, q_{r-2} = S_k, \dots) = P(q_r = S_i | q_{r-1} = S_j)$$

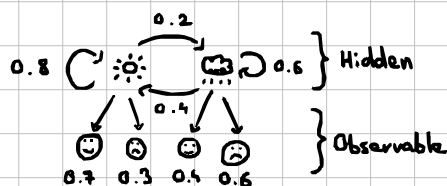
- State transition probabilities:

$$a_{ij} = P(q_r = S_j | q_{r-1} = S_i), \quad a_{ij} \geq 0 \text{ and } \sum_j a_{ij} = 1$$

Hidden states

Say Bob reports his mood based on weather on the phone to someone far.

- Observable states: Bob's mood (happy, grumpy...)
- Hidden states: weather conditions (sunny, rainy...)
 - ↳ the person can only know mood for sure and guess weather from it.



Formal elements of HMM

- N : number of hidden states, denoted as $S = \{S_1, S_2, \dots, S_N\}$
- M : number of observation symbols per state, denoted as $V = \{v_1, v_2, \dots, v_M\}$
- A : state transition probability matrix, denoted as $A = \{a_{ij}\}$ where a_{ij} is $P(q_t = S_j | q_{t-1} = S_i)$
- B : observation symbol probability distribution $B = \{b_i(k)\}$, which defines the probability of emitting v_k in state S_i
- π : initial state distribution $\pi = \{\pi_i\}$, where π_i is the probability of the model starting in state S_i at $t=1$